Problem Chosen	University Physics Competition	Team Control Number
A		758

Controlled Descent and Propellant Estimation for a First-Stage Rocket Booster

Summary

We model the controlled descent and vertical landing of a first-stage booster using coupled translation, pitch, aerodynamic forces, and propellant consumption. Earth curvature is retained through radial altitude, surface arc distance, curvature acceleration, and the changing local horizontal. For the stated 130 t dry mass and a selected 15° separation direction, the reoptimized spherical-Earth trajectory requires 26.04 t of liquid oxygen and methane at separation. This comprises 5.79 t methane and 20.25 t oxygen, including a selected 5 t usable reserve. Approximately 20.96 t is consumed and 5.08 t remains. A conveniently positioned ship lies 284.16 km downrange in surface arc distance. Contact occurs at 213.02 s with horizontal and downward speeds of approximately 0.005 and 1.515 m/s, and local pitch-rate magnitude 0.027°/s. Three replay steps down to 0.00125 s satisfy the adopted position, velocity, attitude, angular-rate, and reserve criteria. Vacuum invariants and symbolic-to-numerical derivative comparisons verify the spherical formulation. The preceding planar design gives 26.23 t and a ship distance of 278.31 km. Planar parameter studies show dependence on engine efficiency, initial direction, and crossflow drag, with one low-crossflow search unresolved. The analysis assumes a rigid booster, a nonrotating Earth, a stationary ship, effective propulsion parameters, and ideal actuators. The inventory is an accepted local design under these assumptions.

Contents

1 Introduction **3**
1.1 Background . . . 3
1.2 Restatement of the Problem . . . 3

2 Assumptions **3**

3 Notations **4**

4 Model Force Analysis **4**
4.1 Gravity and Standard Atmosphere . . . 4
4.2 Axial and Crossflow Aerodynamics . . . 5
4.3 Reference Propulsion Parameters . . . 6

5 Model Motion Analysis **6**
5.1 Momentum Balance and Pitch Dynamics . . . 6
5.2 Spherical-Earth Correction . . . 7
5.3 Initial Conditions and Contact Geometry . . . 8
5.4 Analytic Baselines and Scale Estimates . . . 8
5.5 Numerical Integration and Local Powered-Braking Check . . . 8

6 Landing Strategy and Propellant Estimation **9**
6.1 Search and minimum-propellant formulation . . . 9
6.2 Nominal strategy and separation budget . . . 10
6.3 Dependence on the separation direction . . . 11
6.4 Recomputed Spherical-Earth Reference . . . 13

7 Sensitivity and Flight Diagnostics **13**
7.1 Dependence of the Propellant Budget on Assumptions . . . 13
7.2 Loads Along the Nominal Descent . . . 15
7.3 A Limited Guidance Extension . . . 16
7.4 Interpretation of the Parameter Studies . . . 16

8 Model Evaluation **16**
8.1 P1 Internal Verification and Numerical Convergence . . . 16
8.2 Strengths and Weaknesses . . . 17

9 Discussion and Conclusions **17**
9.1 Attitude Stability and Vertical Touchdown . . . 17
9.2 Load Diagnostics and Scope . . . 18
9.3 Earth Geometry and Rotation . . . 18
9.4 Ship Placement, Wind, and Deck Motion . . . 18
9.5 Conclusions and Practical Recommendations . . . 18

A Reproducibility and Source Code **19**

Report on the Use of AI **21**

1 Introduction

1.1 Background

Reusable first-stage boosters require controlled descent and a safe vertical landing before refurbishment and reuse.

1.2 Restatement of the Problem

At separation, the booster has an altitude of 70 km and a speed of 6000 km/h. The stated dry mass is 130000 kg, its height is 65 m, and its diameter is 7.0 m. It uses liquid oxygen and methane. The recovery ship may be placed at a convenient location. The task is to analyze a safe descent and controlled vertical landing, and estimate the fuel required at separation [11]. The total separation mass is the stated dry mass plus the remaining combined propellant. The problem specifies the initial speed magnitude. The direction affects the time aloft, downrange distance, and state at landing-burn ignition, so it is an explicit scenario input. We use *propellant* for the combined mass of liquid oxygen and methane and *fuel* for the methane component.

2 Assumptions

The booster is initially approximated as a rigid circular cylinder. This external geometry does not establish a uniform internal mass distribution. Atmospheric drag is retained using air-relative velocity and altitude-dependent density. The baseline translational model assumes planar motion and zero wind; steady uniform wind is a subsequent scenario, rather than a justification for neglecting aerodynamic torque.

Aerodynamic moments depend on the offset between the center of pressure and center of mass. A prescribed-axis translational model therefore cannot establish attitude stability or vertical touchdown. Rotational dynamics and control authority must be assessed separately.

The corrected reference uses a nonrotating spherical Earth. Radial altitude and the moving local horizontal enter the dynamics, atmosphere queries, and contact geometry. Initial direction and the unspecified propulsion and aerodynamic parameters are explicit scenario inputs. The booster is rigid, with structural capacity and material failure excluded from the model scope.

The deck is initially stationary and horizontal. Position denotes the booster center of mass; contact geometry must account for vehicle length and, when specified, landing gear and deck height.

Reference Scenario and Mission Conventions

The reference flight-path angle and body pitch at separation are both 15° above horizontal, with zero initial pitch rate. Alternate flight-path scenarios use 0° and 30°, with initial pitch matched to the respective flight-path angle. Ground-relative planar velocity is used. Wind is zero in the nominal case; deterministic horizontal wind scenarios are -10, 0, and $+10$ m/s. These scenario points are analysis choices, not statistical confidence bounds.

The ship is stationary, the deck is horizontal at sea level, and landing gear is omitted. A nominal ship position may be chosen separately for each nominal scenario. Perturbation

tests for an already positioned ship retain that ship location. The selected touchdown tolerances are 5 m horizontal position error, 1 m/s horizontal speed, 2 m/s vertical speed, 2° tilt from upright, and 1°/s pitch rate.

The initial inventory includes a minimum usable touchdown reserve of 5000 kg; unusable residual is assumed zero in the first model. The 80000 kg trial inventory is an initialization, not a fuel requirement. Consumed propellant, reserve, and any surplus are accounted for without double counting. At an oxygen-to-methane mass ratio of 3.5, methane represents 1/4.5 of total propellant mass.

At most three ignitions are allowed as a reference operational assumption. Ignition delay and minimum on/off durations are zero, and actuator rate limits are omitted. Throttle and gimbal changes are therefore idealized as instantaneous; this is not a manufacturer capability claim. No independent reaction-control system or active aerodynamic fins are assumed. Engine-off attitude motion is passive.

Sensitivity cases vary effective specific impulse over 300, 330, and 360 s; axial coefficient over 0.4, 0.8, and 1.2; crossflow coefficient over 0.8, 1.2, and 1.6; and pressure-center offset over $-0.05L$, 0, and $+0.05L$. These deterministic variations test the reference design assumptions. The booster is rigid. Structural capacity and thermal-protection sizing are excluded, while dynamic pressure and acceleration are reported as trajectory diagnostics.

3 Notations

Symbol	Meaning	Unit
$h, \mathbf{r}$	Altitude and position	m
$\mathbf{v}$	Velocity	m/s
m, m_d, m_p	Total, dry, and propellant mass	kg
T, I_{sp}	Thrust and specific impulse	N, s
ρ, C_D, A	Air density, drag coefficient, reference area	kg/m^3, 1, m^2
γ_0	Initial flight-path angle	rad
q	Dynamic pressure	Pa

4 Model Force Analysis

4.1 Gravity and Standard Atmosphere

We use a local downrange–vertical plane. Position refers to the center of mass (COM), and the pitch angle points from the tail toward the nose. Gravity acts through the COM and has magnitude

$$g(h) = g_0 \left(\frac{R_E}{R_E + h} \right)^2, \qquad g_0 = 9.80665 \text{ m/s}^2, \quad R_E = 6356766 \text{ m}. \tag{1}$$

At 70 km this gives $g = 9.5942$ m/s^2, about 2.17% below g_0. This is a spherical, nonrotating reference approximation, not a latitude-resolved gravity model.

Atmospheric properties follow the lower layers of the 1976 standard atmosphere [3]. Geometric altitude h is converted to geopotential altitude $H = R_E h/(R_E + h)$. In a layer

with base (H_b, T_b, p_b) and temperature gradient λ,

$$T_a = T_b + \lambda(H - H_b), \tag{2}$$

$$p_a = \begin{cases} p_b(T_a/T_b)^{-g_0/(R_a\lambda)}, & \lambda \neq 0, \\ p_b \exp[-g_0(H - H_b)/(R_a T_b)], & \lambda = 0, \end{cases} \tag{3}$$

$$\rho = \frac{p_a}{R_a T_a}, \qquad R_a = 287.05287 \text{ J/(kg K)}. \tag{4}$$

These expressions follow from hydrostatic balance $dp_a/dH = -\rho g_0$ and the ideal-gas law. The implemented range ends at $H = 84.852$ km; trajectories outside that range are rejected rather than assigned extrapolated density.

4.2 Axial and Crossflow Aerodynamics

The cylindrical approximation gives $A_{end} = \pi D^2/4 = 38.4845$ m^2 and side area $DL = 455$ m^2. Define body directions

$$\mathbf{e} = (\cos\theta, \sin\theta), \quad \mathbf{n} = (-\sin\theta, \cos\theta), \quad \mathbf{u} = \mathbf{v} - \mathbf{w}, \quad u_a = \mathbf{u}\cdot\mathbf{e}, \quad u_n = \mathbf{u}\cdot\mathbf{n}. \tag{5}$$

Instead of applying one coefficient to a changing projected area, we separate axial and crossflow contributions. Cylinder research motivates treating crossflow as a distinct aerodynamic regime [2]; our constant coefficients are low-order engineering assumptions, not a fit to that report.

The axial force is

$$F_a = -\tfrac{1}{2}\rho C_a A_{end} u_a |u_a|. \tag{6}$$

For an equivalent crossflow strip at signed axial arm s, rotation adds the local normal velocity ωs. Thus

$$dF_n = -\tfrac{1}{2}\rho C_n D(u_n + \omega s)|u_n + \omega s|\, ds. \tag{7}$$

With N midpoint strips of length $\Delta s = L/N$ and $s_i = -L/2 + (i - 1/2)\Delta s + \Delta l$,

$$\mathbf{F}_D = F_a\mathbf{e} + \sum_i F_{n,i}\mathbf{n}, \qquad M_a = \sum_i s_i F_{n,i}. \tag{8}$$

The shifted equivalent distribution represents an uncertain pressure-center offset Δl; it is not a physical surface mesh. We evaluate density at the COM and neglect its variation across the 65 m body. Axial force acts along the centerline. Figure 1 defines the forces and signs.

For zero rotation, $M_a = \Delta l F_n$. Consequently, even spatially uniform wind can generate a moment when $\Delta l \neq 0$. For a centered distribution and pure rotation, direct integration gives

$$M_a = -\frac{\rho C_n D L^4}{64}\omega|\omega|. \tag{9}$$

This opposes rotation. More generally, the mechanical power transferred by the modeled air-relative forces satisfies

$$\mathbf{F}_D \cdot \mathbf{u} + M_a\omega = -\tfrac{1}{2}\rho C_a A_{end}|u_a|^3 - \tfrac{1}{2}\rho C_n D \sum_i \Delta s|u_n + \omega s_i|^3 \leq 0. \tag{10}$$

This identity provides a useful sign and implementation check. It concerns motion relative to air, not ground-frame energy in moving wind.

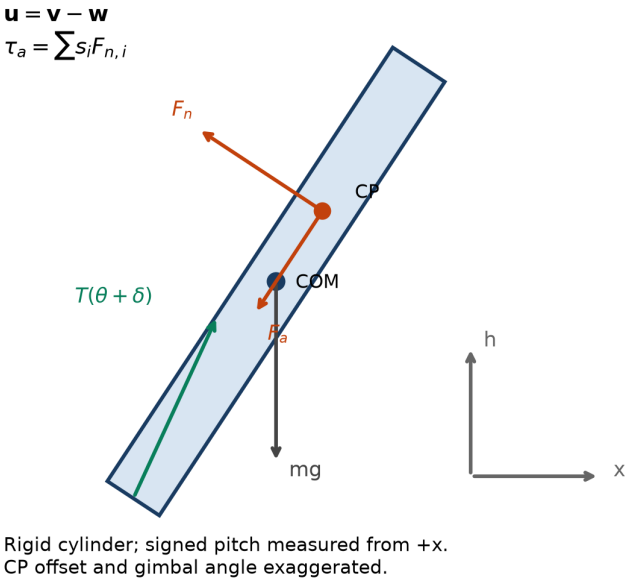


Figure 1: Planar reference geometry and force conventions. The pressure-center displacement is exaggerated. Positive gimbal angle at the tail produces a negative pitch moment.

4.3 Reference Propulsion Parameters

The maximum landing thrust is set to 2.846 MN by analogy with the published BE-4 thrust scale [1]. We select effective $I_{sp} = 330$ s, oxygen-to-methane mass ratio 3.5, minimum throttle 0.4, and gimbal limit 10° as reference-design assumptions. The mixture-ratio choice is informed by a methane/oxygen mission study [4]; that study's vacuum specific impulse is not assigned to this landing engine. We set $C_a = 0.8$, $C_n = 1.2$, and $\Delta l = 0.05L$. The P0 reference contract records exploratory scenario points. We report selected efficiency and crossflow comparisons and retain explicit limitations for unexamined assumptions.

The standard effective-exhaust-velocity definition gives $c = I_{sp} g_0$ and

$$\dot{m} = -T/c, \qquad T = 0 \quad \text{or} \quad 0.4T_{max} \le T \le T_{max}. \tag{11}$$

This consumes both oxygen and methane. Pressure dependence and actuator transients are represented only through the effective parameter assumptions at this stage.

5 Model Motion Analysis

5.1 Momentum Balance and Pitch Dynamics

The state is $\mathbf{y} = (x, h, v_x, v_h, m, \theta, \omega)$. Accounting for exhaust momentum through thrust, the open-system translational balance is $m\dot{\mathbf{v}} = m\mathbf{g} + \mathbf{F}_D + \mathbf{F}_T$. One must not add a second $\dot{m}\mathbf{v}$ term to this thrust-based equation. The planar baseline used to initialize the

corrected calculation is

$$\dot{x} = v_x, \qquad \dot{h} = v_h, \tag{12}$$

$$\dot{v}_x = \frac{F_{D,x} + T\cos(\theta+\delta)}{m}, \qquad \dot{v}_h = -g(h) + \frac{F_{D,h} + T\sin(\theta+\delta)}{m}, \tag{13}$$

$$\dot{m} = -\frac{T}{g_0 I_{sp}}, \qquad \dot{\theta} = \omega, \tag{14}$$

$$I_d\dot{\omega} = M_a - \frac{L}{2}T\sin\delta, \qquad I_d = \frac{m_d(3R_b^2 + L^2)}{12}. \tag{15}$$

The thrust application point is $-(L/2)\mathbf{e}$, so its cross product with thrust gives the negative sign above. The dry body is modeled as a uniform cylinder. Remaining propellant is idealized as concentrated at the COM with negligible pitch inertia and no additional rotational angular-momentum flux. Hence I_d is constant. A distributed-tank model would require consistent treatment of changing inertia and exhaust angular momentum; the present equation does not claim to describe those effects.

5.2 Spherical-Earth Correction

The adopted correction uses a nonrotating spherical Earth with radius R_E. Write $r = R_E + h$ and $s = R_E\phi$, where s is surface arc distance. Velocities v_x and v_h are physical components along the changing local horizontal and radial directions. The state retains the seven-component layout, with x replaced by s. Let $a_x = F_x/m$ and $a_h = F_h/m$ denote non-gravitational accelerations. Differentiating the moving basis gives

$$\dot{s} = \frac{R_E}{r}v_x, \qquad \dot{h} = v_h, \tag{16}$$

$$\dot{v}_x = a_x - \frac{v_x v_h}{r}, \qquad \dot{v}_h = a_h - g(h) + \frac{v_x^2}{r}. \tag{17}$$

Thus surface distance differs from the integral of local horizontal speed, and both acceleration components acquire curvature terms.

We retain $\omega = \dot{\theta}$ as pitch rate relative to the local horizontal. As that horizontal rotates at $-v_x/r$ in the flight plane, inertial body rate is $\omega_I = \omega - v_x/r$. The strip velocity in Eq. (8) uses $u_n + \omega_I s_i$. The consistent angular equation is

$$\dot{\omega} = \frac{M_a + M_T}{I_d} + \frac{a_x}{r} - \frac{2v_x v_h}{r^2}. \tag{18}$$

The last two terms are obtained by differentiating v_x/r. This distinction preserves torque-free inertial rotation while allowing the local vertical to change along the trajectory.

For sea-level contact, the cylindrical longitudinal section is a rectangle in the local flight plane. The point nearest the Earth center has body coordinates

$$a_* = \mathrm{clip}(-r\sin\theta, -L/2, L/2), \qquad b_* = \mathrm{clip}(-r\cos\theta, -R_b, R_b). \tag{19}$$

The event function is

$$f_s = \sqrt{(a_*\cos\theta - b_*\sin\theta)^2 + (r + a_*\sin\theta + b_*\cos\theta)^2} - R_E - h_{deck}. \tag{20}$$

It uses the radial surface and the full cylinder section. Ship position is a fixed surface arc coordinate. The 5 m targeting tolerance keeps the local deck orientation consistent with the touchdown checks to the precision relevant here. Deck motion and Earth rotation remain excluded assumptions.

5.3 Initial Conditions and Contact Geometry

For each selected initial flight-path angle,

$$x_0 = 0, \quad h_0 = 70000 \text{ m}, \quad (v_{x,0}, v_{h,0}) = \frac{6000}{3.6}(\cos\gamma_0, \sin\gamma_0), \quad m_0 = m_d + m_{p,0}. \tag{21}$$

The reference uses $\gamma_0 = \theta_0 = 15^\circ$ and $\omega_0 = 0$. A trial $m_{p,0} = 80000$ kg initializes calculations; it is not a computed minimum.

In the planar comparison, for a cylindrical body with no landing gear, the vertical support extent below the COM is $(L/2)|\sin\theta| + R_b|\cos\theta|$. First deck contact therefore occurs when

$$f(\mathbf{y}) = h - h_{deck} - \frac{L}{2}|\sin\theta| - R_b|\cos\theta| = 0. \tag{22}$$

The spherical calculation instead locates first contact with Eq. (20). An upright body contacts at $h - h_{deck} = L/2$. Speed, attitude, pitch-rate, propellant reserve, and position relative to the chosen ship must be assessed separately. The local diagnostic below checks the former quantities, not fixed-ship position capture.

5.4 Analytic Baselines and Scale Estimates

For constant gravity, zero thrust, and zero drag,

$$h(t) = h_0 + v_{h,0}t - \tfrac{1}{2}g_0t^2, \qquad x(t) = v_{x,0}t. \tag{23}$$

At 15°, $(v_{x,0}, v_{h,0}) = (1609.88, 431.37)$ m/s. The constant-g_0 apex estimate is 79.49 km after 43.99 s. This is an analytic scale estimate, not the integrated reference trajectory. In the variable-gravity vertical vacuum limit, the conserved specific energy is

$$E = \tfrac{1}{2}v_h^2 - \frac{g_0R_E^2}{R_E + h}. \tag{24}$$

Without opposing external acceleration in the thrust direction, integrating $dv = -c\,dm/m$ yields

$$\Delta v = c\ln(m_0/m_f). \tag{25}$$

For 210 t initially and 135 t finally, this gives 1429.86 m/s. It is ideal propulsive capability, not a required landing velocity increment or proof of feasibility. Atmospheric drag may remove speed while gravity and steering alter the propulsion requirement.

With mass frozen, no drag, and vertical maximum thrust, a useful stopping-distance estimate is

$$d_{stop} \simeq \frac{v_h^2}{2(T_{max}/m - g_0)}, \qquad T_{max} > mg_0. \tag{26}$$

For 180 t and a 90 m/s descent, it is 674.5 m. Mass loss and drag are omitted from this estimate. The maximum-thrust weight ratio is 1.382 at 210 t and 2.232 at dry mass, showing why instantaneous mass must enter the dynamics.

5.5 Numerical Integration and Local Powered-Braking Check

A fixed-step fourth-order Runge–Kutta integrator advances the coupled equations. Step size is reduced near the deck, and contact is located by linear interpolation across a sign

change of Eq. (22). This does not imply fourth-order event accuracy. A conservative propellant guard stops before a step could cross dry mass; it prevents negative inventory but does not return the exact depletion time.

To test the powered and rotational implementation independently of whole-flight optimization, we start a local case at $h = 1200$ m, $v_h = -90$ m/s, $v_x = 0$, and $m = 180000$ kg. Define height above upright contact $z = h - L/2$ and a descent-speed reference

$$v_d = -\min\{90, \sqrt{2k_z \max(z, 0.01) + v_c^2}\}, \quad k_z = 1.5 \text{ m/s}^2, \quad v_c = 1 \text{ m/s}. \tag{27}$$

A vertical acceleration command $a_c = k_v(v_d - v_h)$, $k_v = 4\,\text{s}^{-1}$, supplies the throttle request $[m(g_0 + a_c) - F_{D,h}]/T_{max}$, clipped to the permitted operating range. This vertical-axis test controller uses g_0 as an approximation and does not provide general two-dimensional guidance. Pitch control requests

$$M_{req} = I_d[k_p(\pi/2 - \theta) - k_d\omega] - M_a, \quad \delta = \arcsin\left[\text{clip}\left(-\frac{M_{req}}{(L/2)T}, -1, 1\right)\right], \tag{28}$$

where $k_p = 0.5$ s^{-2}, $k_d = 1.2$ s^{-1}, followed by gimbal clipping. With no saturation, this gives the stable error equation $\ddot{e} + k_d\dot{e} + k_p e = 0$. Saturation and engine-off motion invalidate that linear closed-loop conclusion and must be checked in the full trajectory.

At a 0.005 s step the nominal case contacts after 35.97505 s with $v_h = -1.50291$ m/s and consumes 22679.12 kg. A separate test with initial pitch 89° and initial rate 0.005 rad/s ends with approximately zero pitch error and $v_x = 0.4671$ m/s. Its horizontal drift is 16.56 m, exceeding the 5 m fixed-ship position tolerance if the ship remains at $x = 0$. Thus attitude recovery is demonstrated locally, but fixed-ship capture is not. Figure 2 shows these implementation checks; they do not estimate separation fuel demand.

6 Landing Strategy and Propellant Estimation

The planar model is used first to construct feasible seeds and compare strategies. The following planar tables document that baseline. The spherical-Earth result in the final subsection supplies the corrected reference fuel and ship position.

6.1 Search and minimum-propellant formulation

We first propagated the P1 equations with altitude-triggered ignition and a feedback law. During the burn, the pitch target blends a biased retrograde direction into an upright target with lateral-velocity feedback. The throttle follows a decreasing vertical-speed profile and the gimbal supplies a pitch-tracking moment. A search over ignition altitude, propellant inventory, pitch bias, and the vertical-speed profile furnished feasible seeds for one and two ignitions.

We then minimized $m(0) - m_d$ over the phase durations, throttle and gimbal commands, and trajectory states. The phase interfaces share all seven P1 states. Coast phases have zero thrust. Each powered phase obeys $0.4 \le u_T \le 1$ and $|\delta| \le 10°$. The separation state follows P0, with initial mass free. The endpoint uses cylindrical contact geometry, P0's deck-relative velocity and attitude tolerances, and a minimum 5 t usable reserve. The nominal ship position is a free downrange coordinate and is fixed at the selected position during replay. Interior target values of $v_x = 0$, $v_h = -1.5$ m/s,

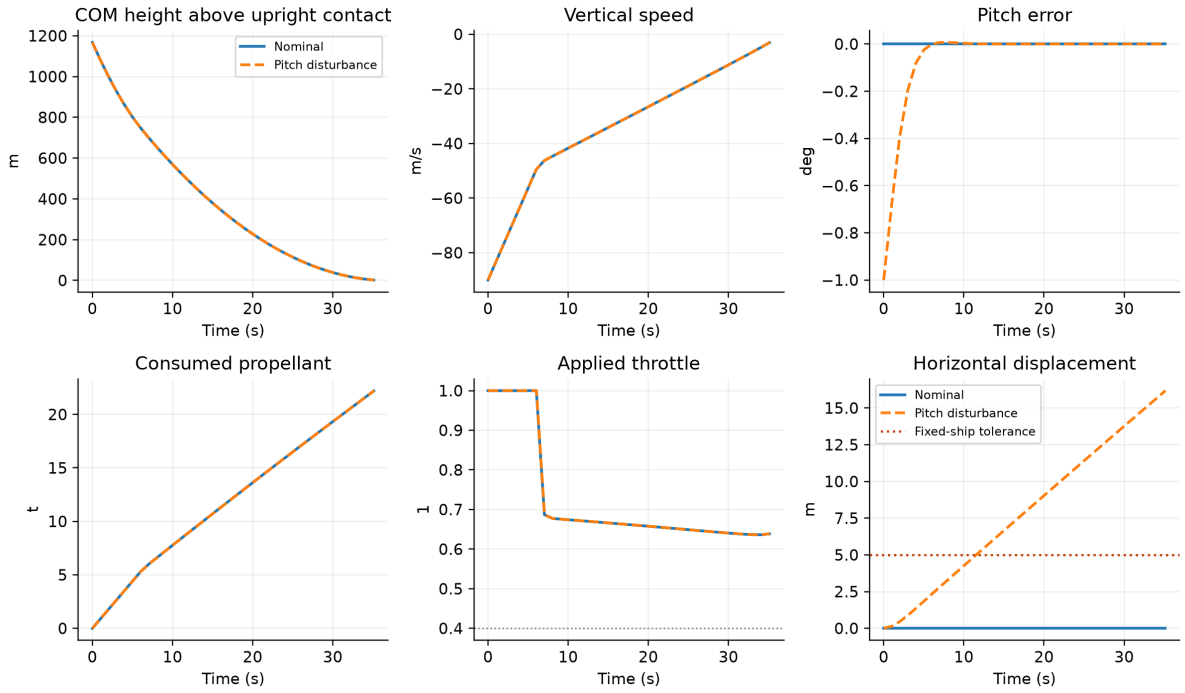


Figure 2: Local powered-braking verification, not a separation-to-landing solution. The disturbed case starts at 89° pitch with 0.005 rad/s pitch rate. The 5 m line highlights that position capture is not guaranteed by attitude recovery.

upright pitch and zero pitch rate, and a numerical reserve margin of 75 kg left room for discretization error.

We initialized Hermite–Simpson direct collocation from the feedback solutions [5]. We refined the candidates using RK4 multiple shooting with eight substeps per control interval and a local trust region in phase duration and pitch. CasADi supplied derivatives to IPOPT [6, 7]. We accepted a candidate only if replay with the original P1 code located the first deck contact and passed the full P0 touchdown checks at both 0.01 s and 0.005 s. Throttle and gimbal were interpolated separately within each phase. The calculation used the specified instantaneous actuator model.

6.2 Nominal strategy and separation budget

For the 15° reference separation in still air, the selected trajectory uses one ignition. The booster coasts for 176.68 s and ignites at a center-of-mass altitude of 5.876 km, while moving downrange at 669 m/s and descending at 580 m/s. The landing burn lasts 30.79 s and reduces both velocity components while bringing the body upright. First deck contact occurs at 207.47 s, with the stationary ship placed 278.310 km downrange.

With that ship position held fixed in the replay, the contact point is within 1 mm of the target. The deck-relative horizontal and vertical velocities are -0.011 and $-1.533\,\mathrm{m/s}$, with magnitudes below the 1 and 2 m/s limits. The body is within 0.001° of upright and the pitch-rate magnitude is 0.059°/s, below the respective limits of 2° and 1°/s. The usable propellant remaining is 5.078 t, above the 5 t requirement. Together, these results establish touchdown feasibility for the reference case under the P0 criteria.

The separation inventory follows from the mass balance

$$m_{p,0} = m_{\text{burned}} + m_{\text{reserve}} + m_{\text{residual}} + m_{\text{surplus}}. \tag{29}$$

The planned usable reserve is 5 t, the unusable residual is set to zero, and surplus is usable propellant remaining above the planned reserve. In the one-ignition case, the coast consumes no propellant. The 21.1505 t burned during landing and the 5.0776 t retained at contact give a separation load of 26.2281 t. The 5 t reserve is already included in that load. With the assumed oxygen-to-methane mass ratio of 3.5, methane accounts for 1/4.5 of the load, or 5.828 t. Liquid oxygen accounts for the remaining 20.400 t.

Table 1: Nominal 15° strategies that satisfy the P0 touchdown criteria in P1 replay. Each strategy uses its own selected ship position. Masses are in metric tonnes and rounded independently.

Quantity	One ignition	Two ignitions
Separation propellant	26.228	27.607
Consumed propellant	21.151	22.516
Planned usable reserve	5.000	5.000
Unusable residual	0.000	0.000
Usable surplus at contact	0.078	0.091
Methane at separation	5.828	6.135
Liquid oxygen at separation	20.400	21.472
Peak dynamic pressure (kPa)	297.3	295.7
Peak proper acceleration (g_0)	7.70	7.70

The accepted two-ignition solution in Table 1 carries 1.38 t more than the one-ignition solution. Its first pulse lasts approximately 0.001 s, at the numerical duration floor, and burns only 0.36 kg. A separate feedback candidate with a 1.5 s entry burn also satisfies the touchdown criteria and requires 38.65 t. The one-ignition strategy therefore gives the lowest accepted inventory among the schedules found for the reference case. This comparison is limited to the candidates returned by the search and the assumed actuator model.

Rerunning the optimization with a nominal shooting interval of 0.55 s instead of 0.85 s reduces the selected inventory by about 70 kg. For the selected schedule held fixed, reducing the replay step from 0.01 s to 0.005 s changes the consumed mass by 0.72 kg. A further reduction to 0.0025 s changes it by 0.15 kg and preserves touchdown feasibility. The replay comparisons isolate the effect of integrating the same controls at different time steps.

For the selected trajectory, dynamic pressure reaches 297.3 kPa at 9.69 km altitude, before ignition. Proper acceleration reaches $7.70 g_0$ at 2.89 km during the burn. These peaks describe the modeled flight environment. Assessing vehicle feasibility also requires structural limits, a heating model with thermal limits, and a model of actuator response during the large pitch excursions.

6.3 Dependence on the separation direction

The problem specifies the speed at separation but leaves its direction open. We compare initial flight-path angles of 0°, 15°, and 30° above horizontal, with the body pitch initially

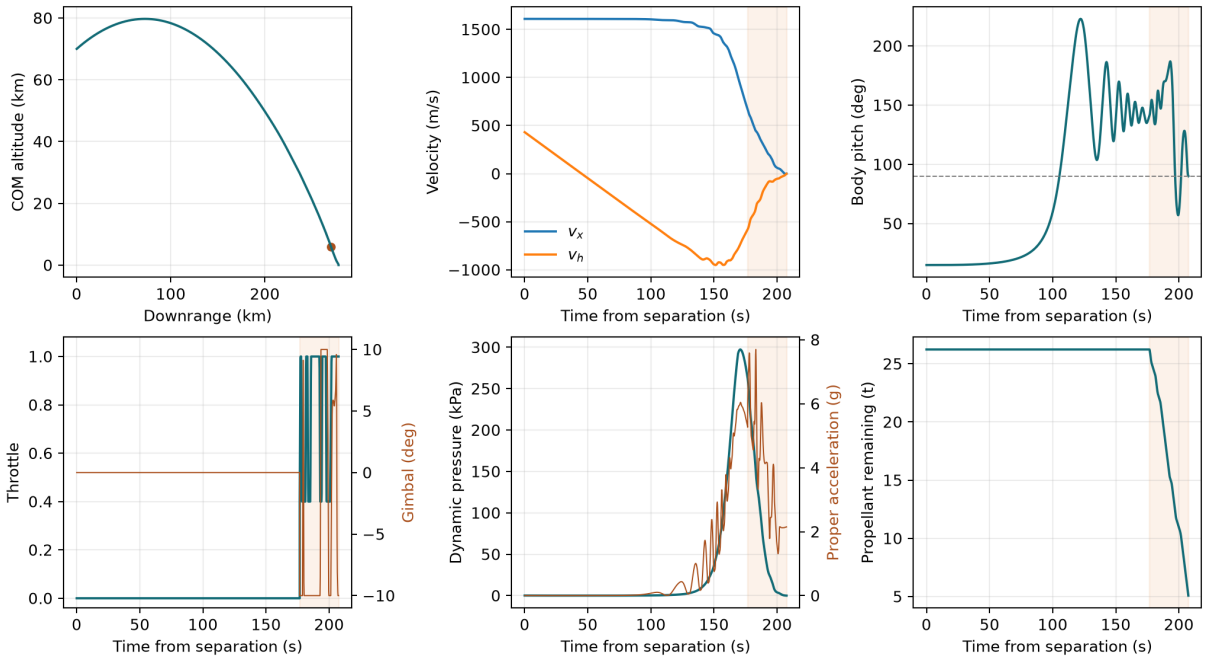


Figure 3: Selected one-ignition trajectory at 15°. The dot and shaded interval mark ignition and the landing burn, respectively. Dynamic pressure peaks before ignition, while proper acceleration peaks during the burn.

aligned with the flight path in each case. We use the same propulsion assumptions, touchdown criteria, and 5 t usable reserve requirement. For each angle we select a recovery site and hold that site fixed when replaying the candidate. Table 2 reports the lowest inventory found to satisfy the touchdown criteria for each case.

Table 2: Accepted strategies for the three separation angles. Ship position is measured downrange from the separation point. The separation load includes the planned 5 t reserve.

Angle	Powered phases	Ship position (km)	Separation load (t)
0°	Landing burn	196.00	24.71
15°	Landing burn	278.31	26.23
30°	Ascent braking and landing burn	260.00	67.00

The 0° and 15° solutions both coast to a single landing burn. The 15° trajectory first climbs to 79.71 km and ignites about 49.40 s later than the 0° trajectory. Its selected ship position is 82.31 km farther downrange, and its separation load is 1.52 t higher. These differences describe the two optimized trajectories with individually selected ship positions.

At 30°, the coast-first trial reaches the upper boundary of the implemented atmosphere about 22 s after separation, while the booster is still climbing. The accepted feedback trajectory starts a braking burn at separation. The burn lasts about 32.13 s, limiting the subsequent apex to 85.90 km. Of the 67.00 t loaded at separation, approximately 28.25 t is burned during ascent braking, 33.41 t during the landing burn, and 5.34 t remains at contact. The 67.00 t load gives a feasible upper bound for this case within the adopted model. Lower-inventory candidates failed the replay checks for atmospheric coverage or

touchdown, so the available runs establish feasibility at 67.00 t and leave the minimum unresolved.

6.4 Recomputed Spherical-Earth Reference

The preceding planar solutions provide initialization and comparison. We then optimize the coupled spherical equations with the same dry mass, propulsion assumptions, initial local flight-path angle of 15°, and touchdown tolerances. The finest accepted local candidate carries 26.038 t of combined propellant, including the selected 5 t reserve. Its methane and oxygen inventories are 5.786 t and 20.252 t. Contact occurs at 213.016 s, with the ship at a surface arc distance of 284.160 km.

The engine ignites at 181.073 s and a radial COM altitude of 6.350 km. Fine replay gives horizontal speed magnitude 0.0051 m/s, downward radial speed 1.5151 m/s, local tilt 0.00014°, and local pitch-rate magnitude 0.0272°/s. Ship-position error is 0.0033 m. Consumption is 20.962 t and retained usable propellant is 5.076 t. Peak dynamic pressure and specific force are 293.02 kPa and $7.819\,g_0$.

Optimization meshes of 0.85, 0.55, and 0.40 s give inventories of 26.114, 26.051, and 26.038 t. All three candidates pass 0.005 and 0.0025 s spherical replay. The selected candidate additionally passes at 0.00125 s. The corrected inventory is about 0.190 t below the 26.228 t planar comparator. Both the control and trajectory were recomputed, so this difference measures the change between the accepted local designs.

Vacuum checks conserve spherical specific energy and angular momentum over a 100 s coast to 7.5×10^{-9} J/kg and 6.7×10^{-16} relative error. Torque-free inertial attitude remains constant to 5.9×10^{-16} rad. Thirty-two sampled derivative comparisons between the symbolic optimizer and JavaScript dynamics agree within 4.0×10^{-16} scaled discrepancy. These tests verify the moving-frame signs and implementation. The results retain the adopted engine, aerodynamic, atmosphere, and actuator assumptions.

7 Sensitivity and Flight Diagnostics

The parameter comparisons in this section were computed for the planar baseline, before the spherical correction. They retain their role as separate assumption studies. A spherical sensitivity envelope would require replanning each case with the corrected dynamics. The propellant estimate depends on quantities not fully specified by the problem. We examine effective engine efficiency and crossflow drag while retaining the reference initial direction and 5 t planned reserve. Each known-parameter design selects a convenient recovery site, as permitted by the problem, and holds that site fixed during replay. These separate designs do not establish recovery from unknown parameter errors at one prepositioned ship. The larger influence of the unspecified separation direction is reported in Section 6.3.

7.1 Dependence of the Propellant Budget on Assumptions

7.1.1 Effective Specific Impulse

We compare effective specific impulses of 300, 330, and 360 s, with maximum thrust, minimum throttle, gimbal limits, aerodynamic coefficients, and oxygen-to-methane mass ratio unchanged. Each changed case reoptimizes inventory, phase durations, and controls

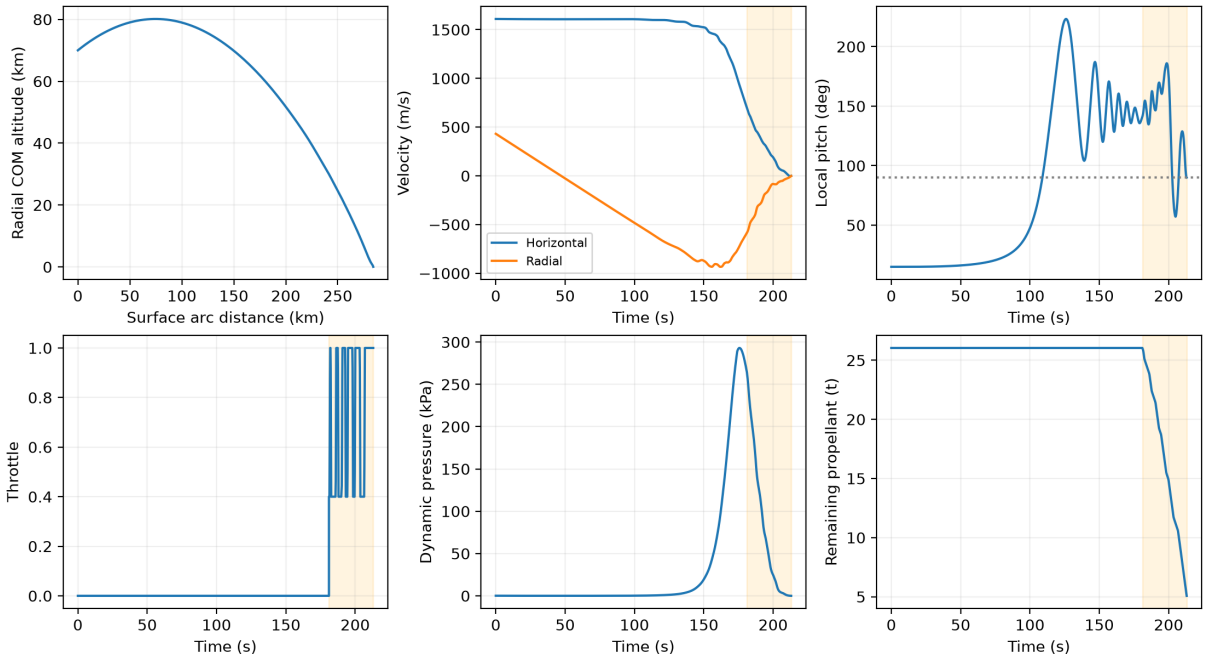


Figure 4: Reoptimized nonrotating spherical-Earth trajectory. Downrange position is surface arc distance, altitude is radial, and pitch is relative to the changing local horizontal. Shading indicates the powered phase.

within the P2 single-ignition family. The 330 s row uses the selected nominal result. All three candidates satisfy the adopted touchdown conditions in full-flight replay at steps of 0.005 s and 0.0025 s; Table 3 uses the finer replay. The candidates are local solutions, not proven global minima.

Table 3: Effective-Isp budgets in tonnes, including at least 5 t retained propellant. Methane and oxygen are components of the separation inventory. Values are rounded independently.

I_{sp} (s)	Inventory	Consumed	Retained	Methane	Oxygen
300	28.781	23.701	5.079	6.396	22.385
330	26.228	21.150	5.078	5.828	20.400
360	24.199	19.124	5.075	5.378	18.821

Relative to 330 s, accepted inventory rises by 2.552 t (9.73%) at 300 s and falls by 2.029 t (7.74%) at 360 s. Higher specific impulse reduces consumption at a prescribed thrust, but total inventory also includes the retained amount, and the changed mass affects the descent. Replanning the complete trajectory therefore differs from simply scaling inventory inversely with specific impulse. The range 24.199–28.781 t describes these known-efficiency designs, not a confidence interval. Holding thrust and mixture ratio fixed is an effective-parameter approximation rather than a thermochemically coupled engine redesign.

7.1.2 Crossflow Drag

Changing the crossflow coefficient affects aerodynamic braking, pitching moment, and rotational damping together. With the axial coefficient fixed at 0.8, we compare crossflow coefficients of 0.8, 1.2, and 1.6 using the same local optimization method and touchdown requirements.

Table 4: Crossflow comparison. Inventories are assigned only to accepted candidates. The unresolved row is a search limitation, not proof that landing is physically impossible.

C_n	Outcome	Inventory (t)	Retained (t)
0.8	No accepted candidate	–	–
1.2	Nominal reference	26.228	5.078
1.6	Accepted local candidate	27.130	5.072

At $C_n = 1.6$, inventory increases by 0.901 t (3.44%), while peak specific force decreases from approximately $7.703\,g_0$ to $7.187\,g_0$. This local result trades greater inventory for a lower peak load; larger crossflow drag does not automatically imply lower fuel demand. At $C_n = 0.8$, the local solve reaches its 140 s limit without an accepted trajectory. We assign no fuel requirement to its failed candidate and do not infer a feasible propellant range across the full coefficient interval.

7.2 Loads Along the Nominal Descent

Safe descent requires considering the flight before touchdown. We use the same 26.228 t nominal trajectory as the fuel calculation, so the load discussion and budget describe one design. Dynamic pressure is $q = \rho U^2/2$, with air-relative speed U [8]. The non-gravitational specific-force magnitude is $\|\mathbf{F}_a + \mathbf{T}\|/(mg_0)$; it differs from ground-relative acceleration, which includes gravity.

Table 5: Nominal-trajectory load peaks, extracted at a 0.0025 s replay step. Height denotes centre-of-mass altitude.

Quantity	Time (s)	Height (km)	Peak
Dynamic pressure	170.837	9.694	297.334 kPa
Specific-force magnitude	183.126	2.895	$7.703\,g_0$
Normal aerodynamic force	183.221	2.861	9.615 MN

Maximum dynamic pressure precedes ignition at 176.68 s, whereas the largest specific force occurs during the burn. Near the force peak, the body axis makes an acute angle of about 20° with the air-relative velocity line, producing substantial normal loading. Thus attitude matters as well as speed and density. Reducing a pre-ignition load peak requires changing the preceding descent or adding earlier braking; a later correction alone cannot remove it. Such a change would also alter the propellant budget.

The table describes the preceding planar baseline. The corrected spherical trajectory has peak dynamic pressure 293.021 kPa and peak non-gravitational specific force $7.819\,g_0$. These quantities characterize the rigid-body flight environment. Structural capacity and thermal-protection sizing are outside the adopted scope.

7.3 A Limited Guidance Extension

Holding the nominal commands fixed does not generally preserve touchdown under wind or initial-attitude errors. An illustrative 30 t feedback design recovered five tested steady-wind errors, -0.1, -0.05, 0, $+0.05$, and $+0.1$ m/s, at the same ship under exact state information and instantaneous actuator response. Its nominal peak specific force was approximately $8.13\,g_0$, above the reference value. This supplementary calculation demonstrates local corrective action, not a general recovery envelope. The trial inventory is not a replacement for the nominal propellant estimate. Detailed replanning, wind-error sweeps, and controller settings are retained in the repository's supplementary calculation records.

7.4 Interpretation of the Parameter Studies

The planar comparisons vary known parameters in separate designs, each with its own accepted trajectory and recovery location. The selected 5 t reserve is included in every reported inventory. The unresolved low-crossflow search prevents assigning a feasible fuel interval across the complete coefficient range. The corrected spherical reference in Section 6 is the current nominal answer. The supplementary planar feedback experiments remain examples of local error recovery under their stated conditions.

8 Model Evaluation

8.1 P1 Internal Verification and Numerical Convergence

The verification suite tests implementation consistency and known limiting solutions; no flight-data validation is claimed. Standard-atmosphere base pressures at 0, 11, 20, 32, and 47 km geopotential altitude agree with rounded reference values within 4.8×10^{-6} relative error. Out-of-range atmospheric queries are rejected. A vertical vacuum coast conserves the modeled specific mechanical energy to 7.5×10^{-9} J/kg over 20 s. Constant-thrust mass history and horizontal ideal-rocket velocity agree with their analytic expressions to 7.6×10^{-10} kg and 2.9×10^{-14} m/s, respectively, in the chosen tests. These small residuals apply to those tests and do not measure real-world accuracy.

Pure axial and pure crossflow force limits, static pressure-center torque, torque-free rotation, damping sign, gimbal sign, and air-relative dissipativity are checked. For pure centered rotation, Eq. (9) provides an independent integral. Midpoint-strip relative errors are 2%, 0.5%, 0.125%, 0.03125%, and 0.0078125% for 10, 20, 40, 80, and 160 strips. Thus the default 20-strip rotational torque has a 0.5% quadrature error in this limiting case; this is distinct from uncertainty in the aerodynamic coefficients.

A four-second smooth unpowered trajectory is compared with a 0.00125 s reference using the same equations. Position differences decrease from 9.75×10^{-7} m at 0.08 s to 2.76×10^{-10} m at 0.01 s. Since force laws contain absolute values and feedback contains clipping, fourth-order behavior is not assumed across every nonsmooth transition. Figure 5 separates smooth integration, strip quadrature, and touchdown-event checks. Between the final two steps, consumption changes by 0.00384 kg and contact time by 1.60×10^{-5} s. Event speed differences are not monotonic, so no fourth-order contact-convergence claim is made. The fuel-depletion guard is verified to avoid dry-mass overshoot, while exact depletion-time localization remains outside this conservative

Table 6: Local terminal-event convergence. All rows satisfy the selected speed, attitude, and reserve criteria; this table does not test fixed-ship position capture.

Step (s)	Contact time (s)	v_h (m/s)	Consumed (kg)
0.080	35.975945	-1.502940	22679.544
0.040	35.975286	-1.503756	22679.191
0.020	35.975097	-1.502978	22679.139
0.010	35.975063	-1.502986	22679.123
0.005	35.975047	-1.502909	22679.119

guard's contract. The reference implementation and P1 checks are reproducible with

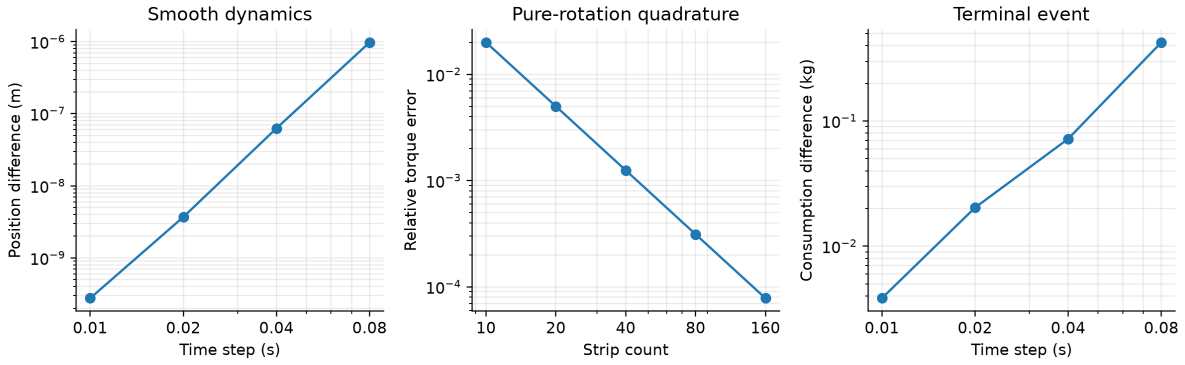


Figure 5: Independent numerical error sources. Left: smooth-trajectory position difference relative to a finer same-model integration. Middle: strip torque error against the analytic integral. Right: contact-consumption difference relative to the 0.005 s run.

`node src/verify-p1.mjs`. The remaining limitations are model-form uncertainty, actuator idealization, atmosphere range, local-plane/rotation approximations, and absent structural/thermal allowables. Whole-flight feasibility and fuel optimization are delegated to the next stage.

8.2 Strengths and Weaknesses

The model accounts for changing mass and pitch motion while locating the first contact of the tilted body with the deck. This makes the landing state and propellant budget mutually consistent. Its principal limitations are assumed propulsion and aerodynamic parameters, ideal actuator response, simplified inertia, and omission of Earth rotation. Numerical agreement within this model does not establish the accuracy of those physical assumptions. The parameter comparisons in Section 7 address some of their effect on the answer, within the rigid-body modeling scope.

9 Discussion and Conclusions

9.1 Attitude Stability and Vertical Touchdown

The reference trajectory satisfies position, relative speed, tilt, and pitch-rate conditions at the same contact event. Upright attitude alone would not establish a controlled landing.

During engine-off coast the model has no active attitude actuator; during the burn, correction is limited by thrust and gimbal authority. These restrictions qualify the nominal touchdown result.

9.2 Load Diagnostics and Scope

The booster is treated as a rigid body. Material failure, structural sizing, and thermal-protection design are outside the adopted task scope. Dynamic pressure, aerodynamic moments, and non-gravitational specific force remain trajectory diagnostics. The terminal requirements concern contact position, relative velocity, attitude, angular rate, and propellant reserve. Engine amplitude limits are enforced, while actuator response remains idealized.

9.3 Earth Geometry and Rotation

The nominal solution is recomputed with Eqs. (16)–(20). Surface arc distance, changing local vertical, curvature acceleration, and inertial body rotation are coupled during optimization and replay. The earlier 6.09 km tangent-surface offset served as a reason to replace the geometry. It is not applied as an additive correction to a planar landing point.

Earth is assumed nonrotating in the corrected reference design. Latitude and heading are unspecified. Earth-fixed Coriolis terms would require those inputs and a corresponding three-dimensional trajectory. The present calculation consistently uses the nonrotating spherical model and a stationary recovery site.

9.4 Ship Placement, Wind, and Deck Motion

Downrange placement can avoid a return maneuver. Once the ship is positioned, the guidance must target its location and meet the speed and attitude limits relative to its deck. The present stationary, horizontal deck sets the reference for the fixed-site examples. A moving deck would alter the target position, relative velocity, and contact orientation.

9.5 Conclusions and Practical Recommendations

For the spherical-Earth reference with a 15° separation direction, a coast followed by one landing burn gives an accepted local separation inventory of 26.038 t. At the selected oxygen-to-methane mass ratio of 3.5, this is 5.786 t methane and 20.252 t oxygen. The 5 t usable-reserve requirement is included. The recovery ship is placed at 284.160 km surface arc distance, and the full rigid-body trajectory satisfies the adopted contact constraints.

The corresponding planar design used 26.228 t and a ship distance of 278.310 km. Reoptimization with consistent moving-frame dynamics corrects both the trajectory and its target location. Engine-efficiency, separation-direction, and aerodynamic comparisons from the planar baseline quantify assumption dependence, while a lower-crossflow case remains unresolved. These separate studies do not define a universal inventory for all parameter combinations. The reported solution assumes the reference propulsion and aerodynamics, a nonrotating Earth, a stationary ship, and ideal actuators. Structural capacity is outside the rigid-body study.

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A Reproducibility and Source Code

The corrected reference parameters are in `config/spherical.json`. The planar comparison uses `config/reference.json`.

The touchdown conventions are stored in `config/p0-contract.json`. Spherical optimization and replay are reproduced with `src/spherical-optimize.py` and `src/spherical-report.py`. The accepted controls and finer-step replay are archived in `results/spherical/selected.json`. The selected complete trajectories and inventories are in `results/p2-p0-v1/`; solver commands and dependencies are documented in `docs/p2-landing-strategy.md` and `requirements-p2.txt`.

The efficiency, crossflow, and nominal-load records are in `results/p3/propulsion/`, `results/p3/drag/`, and `results/p3/safety/`. Their drivers are `src/p3-propulsion.`

`py`, `src/p3-drag.py`, and `src/p3-exposure.mjs`, with reproduction commands in the corresponding documentation. The additional fixed-ship replanning and feedback experiments remain in `results/p3/replanned/`, `results/p3/wind-estimation.json`, and `results/p3/tracking/`. These supplementary experiments are distinct from the nominal design used to answer the propellant question.

Report on the Use of AI

AI assistance was used to organize the manuscript, develop the reference dynamics and verification scripts, derive model equations, and generate P1 figures and draft text. It was also used to implement the P2 search, optimization, and replay, generate its figure, and draft the landing-strategy section. OpenAI Codex also assisted with the sensitivity calculations, corrective planning, tracking-controller implementation, flight-environment and propulsion-scenario analysis and figures, and manuscript integration. The computational and writing assistance was provided through OpenAI Codex. Team interactions included requests to select model assumptions, identify parameter sources, implement dynamics, verify trajectories, analyze sensitivity, and integrate the manuscript. The contest requires verbatim queries and complete outputs in this report. Complete interaction records from every participating team member must accompany the final submission. The repository currently contains the computational artifacts and this usage statement. A consolidated verbatim transcript has yet to be supplied by the team.